

## Chapter 11: Work and Energy

Textbook Exercise Solutions & Answers



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### 1. Fill in each blank with the appropriate term from the brackets.

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- (a) A bucketful of water is to be drawn from a well ..... will be done when a ..... is applied to do this, because there will be an ..... of water. (displacement, workforce)

work, force, displacement

- (b) If a ball is dropped on the sloping roof of a house, it acquires ..... and falls on the ground. That is transformation of ..... energy into ..... energy takes place. (kinetic, potential, motion)

motion, kinetic, potential

- (c) You might have seen some beautiful fireworks during Diwali. It is an example of transformation of ..... energy into ..... energy. (light, atom, chemical, solar)

chemical, light

- (d) The solar cooker is an application of the ..... energy of the sun, while solar cells, solar lamps are applications of the ..... energy of the sun. (light, chemical, heat)

heat, light

- (e) One labourer carried four pans of road metal through a distance of 100 metres. If he carries two pans of road metal through a 200 metre distance ..... work will be done. (equal, more, less)

equal

- (f) The capacity that an object has for doing work is called ..... . (energy, displacement, force)

energy

## 2. Match the pairs.

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Column 'A' Column 'B' 1. Rolling object a. Heat energy 2. Food b. Atomic energy 3. Stretched bow c. Kinetic energy 4. Sunlight d. Potential energy 5. Uranium e. Chemical energy

| Column 'A'        | Column 'B'          |
|-------------------|---------------------|
| 1. Rolling object | c. Kinetic energy   |
| 2. Food           | e. Chemical energy  |
| 3. Stretched bow  | d. Potential energy |
| 4. Sunlight       | a. Heat energy      |
| 5. Uranium        | b. Atomic energy    |

## 3. Can you tell?

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### (a) When can we say that displacement has taken place?

When the position of an object changes, we say that the object is displaced.

### (b) What are the various forms of energy?

Various forms of energy are: mechanical energy, heat energy, light energy, sound energy, chemical energy and electrical energy.

### (c) Describe the natural chain of transformation of energy.

1. In the course of water cycle, water evaporates due to the heat of the sun.
2. This water vapour forms clouds that give rain.
3. Rainwater flows into rivers and is stored in reservoirs or dams.
4. This stored water at a height has potential energy.
5. As the water falls down, this is transformed into kinetic energy.
6. When the water falls on the turbine, its kinetic energy is transferred to turbine.
7. The turbine rotates producing electrical energy.

### (d) What should be taken into account for measuring work?

For measuring work both force and displacement must be taken into account.

### (e) Why should we save energy?

Due to increasing population and increasing use of energy resources, there is a danger that limited reserves of coal, petrol, etc. will get exhausted. Hence, we should save energy for future generation.

**(f) What is 'green energy'?**

The energy resources which do not produce smoke and carbon gases such as carbon dioxide or carbon monoxide are called 'green energy resources'. Energy from them is called green energy.

**(g) What are the non-conventional energy resources?**

The resources that:

1. have not been used traditionally.
2. are inexhaustible and continuous.
3. That can be used in various forms again and again are called non-conventional energy resources.
4. e.g.: solar energy, wind energy, tidal energy, hydal power.

**(h) Which forms of energy from the sun is used in solar energy devices?**

1. Heat energy from the sun is used in solar cookers, solar water heaters, solar driers.
2. Light energy from the sun is used in solar cells, solar panels.
3. Solar electric plants have the capacity to produce electricity on large scale.

**(i) Why should we maximize the use of non- ntional energy sources?**

1. Due to increase in population there is increasing use of conventional sources like petrol, coal, etc. Their reserves are limited and they may get exhausted.
2. Hence, it will be better to use non- conventional sources of energy to save conventional resources for future.

**4. Who is the odd-one-out?**

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**(a) Diesel, crude oil, natural gas, wind.**

wind

**(b) A running car, hauling a log, a book kept on a table, picking up the school bag.**

A book kept on table.

**(c) Sunlight, wind, wave, petrol.**

Petrol.

**(d) Leaving the fan on in a vacant room, leaving the TV on while working, using A.C during winter, putting off the light when going out.**

putting off the light when going out.

## 5. Find out the types of energy from the following puzzle.

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Potential, Kinetic, Solar, Wind, Sound, Light, Heat

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|---|---|---|---|---|---|---|---|---|
| z | s | q | p | y | m | w | n | e |
| p | o | t | e | n | t | i | a | l |
| l | l | s | u | h | v | n | x | i |
| t | a | o | j | e | v | d | z | g |
| q | r | u | l | a | b | a | d | h |
| k | i | n | e | t | i | c | q | t |
| r | w | d | h | k | l | w | y | f |